QA TEST EXECUTION REPORT · EAI-QA-TR-001

Client Document Upload Test — Harborview Commons

Application: estate AI (local build) · Executed: 16 Oct 2026, 20:31–20:34 IST · AI model: gemini-2.5-flash · DB: PostgreSQL 17.11
Test input: **Harborview_Commons_Loan_Package.pdf** (12 pages, text-layer PDF) · Target deal: **DL-1001** · Lender policy: Apex Commercial Lending (DSCR ≥ 1.25x, LTV ≤ 75%)

1. Executive Summary

40	35	5	87.5%	6
CHECKS EXECUTED	PASSED	FAILED	PASS RATE	DEFECTS RAISED

Verdict: NO-GO. The upload, extraction, database and audit pipeline work well. But one **critical** defect changes the lending decision. The AI read the **10-year loan term (120 months)** as the amortization period instead of the **30-year amortization (360 months)**. As a result DSCR was calculated as **0.68x instead of 1.20x**, the deal was graded **Weak instead of Marginal**, and the credit memo recommended **Decline**. A borrower with a fundable (conditionally approvable) deal would be wrongly declined.

Expected vs Actual — Key Numbers

Metric	In client PDF	Expected in app	Actual in app	Result
Loan amount	$18,500,000	$18,500,000	$18,500,000	PASS
Interest rate	6.75%	0.0675	0.0675	PASS
Property value	$26,000,000	$26,000,000	$26,000,000	PASS
NOI (T-12)	$1,728,359	$1,728,359	$1,728,359	PASS
Amortization (termMonths)	30 yrs / 360 mo (term 10 yrs)	360	**120**	FAIL
Annual debt service	—	$1,439,888	**$2,549,095**	FAIL
DSCR	—	1.20x	**0.68x**	FAIL
LTV	—	71.15%	71.15%	PASS
Debt yield	—	9.34%	9.34%	PASS
Grade	—	Marginal	**Weak**	FAIL
Memo recommendation	—	Approve w/ Conditions or Decline (1.20x < 1.25x)	Decline (based on wrong 0.68x)	Wrong basis

Expected values come from running the project's own `backend/app/finance.py` on the figures in the PDF.

2. What Worked Well

- Upload accepted (HTTP 200, 53 s). All 12 pages were read from the text layer, so OCR was correctly skipped.
- Document type detected as **Loan Application**. **61 fields** extracted, each with page number and confidence. The summary paragraph was accurate.

- 4 of 5 financial fields extracted exactly. The AI *did* extract `amortization_term_months = 360` as a field but put the wrong value into `financials`.
- 8 discrepancies against the old deal listed (NOI, loan, value, term, borrower, sponsor, property, city).
- Risk flags correctly raised: tax reassessment (+$55–70K/yr), 25.8% lease rollover in Q4 2026, tax returns, bank statements and KYC outstanding.
- Deal re-underwritten automatically. The Inbox updated, and stale underwriting/memo analyses were invalidated and regenerated.
- **Fixed-formula maths held:** the memo quoted exactly the computed DSCR and debt service, and stress tests were consistent (NOI −10% → 0.61x).
- Copilot answered honestly with "Unknown" rather than inventing facts. Chat history (10 messages) saved. All 8 AI calls logged with latency and status.

3. Detailed Test Results

ID	Check	Actual	Result
T01	AI engine + DB healthy	ok=true, model gemini-2.5-flash, database ok	**PASS**
T02	Pipeline loaded, deal selected	10 deals; DL-1001 (Multifamily, no existing debt)	**PASS**
T03	Upload accepted	HTTP 200 in 53.3 s	**PASS**
T04	Text PDF — no OCR	ocr = false	**PASS**
T05	Page count = 12	12	**PASS**
T06	Document type detected	Loan Application	**PASS**
T07	≥ 15 fields extracted	61 fields	**PASS**
T08a–e	Financials extracted (loan, rate, term, value, NOI)	4/5 correct; termMonths = 120 (expected 360)	**1 FAIL**
T09a–e	Deal fields updated	4/5 correct; termMonths = 120	**1 FAIL**
T10a–f	Re-underwritten metrics	LTV, DY, passesDscr, passesLtv correct; DSCR 0.68 (exp 1.20), grade Weak (exp Marginal)	**2 FAIL**
T11	Discrepancies vs deal listed	8 discrepancies	**PASS**
T12	Occupancy conflict (application 96% vs rent roll 95%) flagged	Both values extracted as fields, but **not** flagged as a conflict or risk	**FAIL**
T13	Tax reassessment risk flagged	Flagged ($55K–$70K/yr)	**PASS**
T14	Lease rollover risk flagged	Flagged (25.8% Q4 2026)	**PASS**
T15	Outstanding documents noticed	Tax returns, bank statements, KYC, title, insurance	**PASS**
T16	Document saved & listed	Listed under DL-1001	**PASS**
T17	Inbox shows new metrics	Inbox row matches deal metrics	**PASS**
T18	Underwriting regenerated	8 extractions, 4 risk flags (see DEF-04)	**PASS**
T19	Credit memo generated	Decline	**PASS**
T20	Memo not unconditional Approve	Decline	**PASS**
T21	Memo quotes computed DSCR	0.68x quoted, matching computed value	**PASS**
T22	Stress tests consistent	NOI −10% 0.61x · Rate +100bp 0.61x · Occ 85% 0.58x · NOI +10% 0.75x	**PASS**
T23a–c	Copilot answers 3 questions	All answered (accuracy issue, see DEF-02)	**PASS**
T24	Chat history persisted	10 messages	**PASS**
T25	AI calls logged	document 51.9 s · underwriting 15.9 s · memo 61.5 s · chat 2.3–3.9 s — all ok	**PASS**

> T22 was first scored FAIL by the automated script because its rule wrongly expected the *upside* scenario to be below base. It was re-scored PASS on manual review. T12 was first scored PASS by a loose keyword match. On manual review the conflict was never flagged, so it was re-scored FAIL. The 35/40 total includes both corrections.

4. Defects Raised

S1 CRITICAL **DEF-01 — Loan term used as amortization, so DSCR is wrong**

Observed: PDF states "Loan Term 10 years (120 months) · Amortization 30 years (360 months)". `financials.termMonths` = 120, so debt service is $2,549,095 and DSCR 0.68x instead of $1,439,888 and 1.20x.

Root cause: the upload prompt in `backend/app/main.py` (line ~377) asks only for `"termMonths":number` and never says this means *amortization*. The deal-generation prompt (line 80) does define it as amortization, so the two prompts are inconsistent.

Impact: wrong grade, wrong memo recommendation, wrong Copilot answers. Any loan with a term shorter than its amortization (most CRE loans) is affected.

Suggested fix: define the field explicitly ("*termMonths: AMORTIZATION period in months, NOT the loan maturity term*"), or extract both `loanTermMonths` and `amortizationMonths` and use the latter in `finance.py`.

S2 HIGH **DEF-02 — Copilot cannot see uploaded documents**

Observed: "What documents are still outstanding?" got the answer "rent roll and insurance". The rent roll *was* provided; the real outstanding items are tax returns, bank statements and KYC. "What is the tax reassessment risk?" got the answer "Unknown", even though the document extraction flagged it.

Root cause: the chat prompt (`main.py` lines 240–246) only includes DEAL, UNDERWRITING and MEMO. It does not include `uploaded_documents` (summary, fields, risk flags).

Suggested fix: add each document's summary, riskFlags and key fields to the Copilot context.

S2 HIGH **DEF-03 — Mixed deal identity after upload**

Observed: the AI detected that borrower, sponsor, property and city all differ from the deal, but only the financials were applied. The memo now says "*Apex Multifamily Holdings LLC … The Meridian at Brickell, Miami, FL*" while using Harborview's (Austin) numbers.

Root cause: `main.py` lines 391–401 apply only `financials`. There is no check that the document belongs to the deal, and there is no way to create a new deal for a new client.

Suggested fix: warn or block when identity fields mismatch, and add a "New deal from document" intake flow so a new client can be onboarded.

S2 HIGH **DEF-04 — Underwriting workspace shows simulated sources, not the real document**

Observed: the Extracted Facts panel cites "P&L Statement, FY2023", "Purchase Agreement, Exhibit B" and "Term Sheet, V2". None of these exist in the uploaded package.

Root cause: `underwriting_prompt` (`main.py` line 162) tells the AI to "*simulate what document extraction found*" and does not receive the uploaded documents. This contradicts the README's "No fake data" principle.

Suggested fix: when documents exist, build extractions from the stored `extracted_data` (real source and page); simulate only when no documents are uploaded, and label that clearly.

S3 MEDIUM **DEF-05 — Memo says package 100% complete although documents are outstanding**

Observed: `packageCompleteness = 100` while tax returns, bank statements and KYC are listed as outstanding in the client PDF.

Root cause: `memo_prompt` (line 172) receives no document data. **Suggested fix:** pass document checklists and risk flags into the memo prompt.

S3 MEDIUM **DEF-06 — Internal occupancy conflict not flagged**

Observed: the application states 96% occupancy, while the certified rent roll shows 95.0% (114/120). Both values were extracted but no conflict was raised. **Suggested fix:** add an instruction to the extraction prompt to cross-check values *within* the document, not only against the deal.

5. Copilot Transcript (as tested)

Q: Why does this deal fail lender policy?
A: This deal fails lender policy primarily because the DSCR of 0.68x is significantly below the minimum policy requirement of 1.25x. The proposed annual debt service of $2,549,095 also far exceeds the NOI of $1,728,359…
Assessment: reasoning correct, but the numbers are wrong because of DEF-01.

Q: What documents are still outstanding from the borrower?
A: The borrower still needs to provide the latest rent roll … and evidence of property and liability insurance…
Assessment: incorrect (DEF-02). Expected: 2024–25 tax returns, 3 months bank statements, KYC/OFAC, title and insurance binder.

Q: What is the real estate tax reassessment risk and how much could it reduce DSCR?
A: Unknown. The provided context does not contain information regarding real estate tax reassessment risk…
Assessment: honest (no hallucination), but the information was available in the uploaded document (DEF-02). Expected: +$55–70K taxes → NOI ≈ $1.66M → DSCR ≈ 1.15x.

6. Performance Observed

Operation	Time	Target (Test Plan)	Result
Deal list (cached)	0.1 s	< 0.5 s	**PASS**
Upload + AI extraction (12-page PDF)	53.3 s	< 30 s	**FAIL (AI latency)**
Underwriting regeneration	16.0 s	—	Info
Credit memo generation	61.6 s	—	Info (long; show progress)
Copilot answer	2.4–4.0 s	—	**Good**

7. Recommendation & Next Steps

#	Action	Priority
1	Fix DEF-01 (amortization vs term) and re-run this exact test. Expected DSCR 1.20x, grade Marginal.	S1
2	Feed uploaded document data into Copilot, Underwriting and Memo prompts (DEF-02, 04, 05).	S2
3	Add a client-identity check and a "new deal from document" intake flow (DEF-03).	S2
4	Add an in-document cross-check instruction (DEF-06); show progress text for long AI steps.	S3

Release decision: ☐ Go ☐ Go with conditions ☒ **No-Go until DEF-01 is fixed and retested**.

Evidence: raw API responses saved in `scratch/pdf_test_results.json`; test script `scratch/run_pdf_test.py`. Note: deal DL-1001 in the local database now contains the Harborview figures and the uploaded document.